TRANSPENT

The paper khata, rebuilt for the phone already behind the counter

*"The paper ledger behind the counter has no backup.
If it's lost, so is every balance it ever held."*

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TIMELINE	March 2026 to Present
STATUS	Live · signed release APK, sideloaded via GitHub Releases
STACK	Java · Material Components (MDC 1.12) · AndroidX AppCompat · org.json · Google Drive REST API v3
REPOSITORY	github.com/vinamrapandey/Transpent

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The Problem

Small shop owners across India run credit the same way they have for decades: a paper khata, kept behind the counter, with every customer's running balance and every supplier's pending bill written by hand.

The paper ledger works until it doesn't. Pages tear. Handwriting fades. A customer disputes a balance and there is no entry history to show them. A shop owner short on time miscopies a total from one page to the next. And if the notebook itself is lost, so is the entire record of who owes what.

The obvious digital alternatives do not fit the job. Full accounting software is built for people who already think in ledgers and journals, and it is overkill for an owner who just wants two numbers: what customers owe the shop, and what the shop owes its suppliers. A spreadsheet assumes a laptop most counters do not have. And any cloud first app assumes a signal and a Google account the owner may not want to set up just to open the till.

> *Transpent starts from the paper khata, not from an accounting textbook: the same mental model, kept on the phone that is already behind the counter.*

It keeps the exact structure a shop owner already trusts (a customer side, a supplier side, running balances, dated entries) with no account, no signal, and no learning curve required to open it for the first time.

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The Product

Transpent is a local first Android ledger built around one counter: a small shop, one owner, and two directions of money: what customers owe the shop, and what the shop owes its suppliers.

What a shop owner sees

- A home dashboard with a pill switcher between Customers (emerald green) and Suppliers (sapphire blue): same screen, same actions, a different color so the direction of money is never ambiguous
- Add, Pay, and Hist shortcuts sitting directly on the balance card, with no menu diving to record a sale or a payment
- A Ledger tab listing every customer or supplier as a card, tapping straight into an item entry dialog
- A Products catalog so a repeat item can be added to any customer's tab in two taps instead of retyping name and price each time
- A Search tab surfacing frequent contacts first, then matching customers, suppliers, and products as the owner types
- A History tab of every dated entry, filterable by Customers or Suppliers
- A Statistics screen totalling real customer due and supplier due amounts in matching theme colored cards
- One tap CSV export, either a single combined file or one file per party, ready to hand to an accountant

What Transpent manages underneath

- Party records (customers and suppliers) with name, phone/note, a list of item entries, and any misc lump sum payments
- Entry level detail per item (name, quantity, price, amount paid, date) so a partial payment against one item is tracked precisely, not averaged across the account
- Bill photo attachments (camera or gallery) on supplier records, giving a physical delivery bill a digital backup
- Optional Google Drive backup and restore, entirely opt in. Tapping Continue Locally skips sign in completely and the app is fully usable offline from the very first launch

How it compares

	Transpent	Paper khata	Tally / Vyapar style	WhatsApp / Notes
Works fully offline	Yes	Yes	Partially	Yes
No account required	Yes	N/A	No	Usually no
Separate customer / supplier view	Yes, color coded	Manual	Yes, in menus	No
Survives loss or damage	Optional Drive backup	No	Cloud dependent	Chat history only
Learning curve	Minutes	None	Moderate to high	None, but unstructured
Per item partial payments	Yes	Manual math	Yes	No

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Key Technical Decisions

Decision 1: Programmatic UI, not XML layouts

Every screen in Transpent is built at runtime in Java: zero layout XML files exist for the app's screens. MainActivity.java is the entire UI: dashboard, ledger, search, products, history, statistics, and every dialog. For a solo developer iterating fast on a single activity app, this collapses the edit, compile, and run loop into one file, with no XML and Java ID mismatches to debug. The tradeoff, no Android Studio layout preview, was accepted deliberately in exchange for iteration speed and a codebase a single person can hold in their head.

Decision 2: A local JSON file, not Room or SQLite

Transpent stores its entire ledger (customers, suppliers, products, every entry) in one **ledger.json** file inside the app's private **filesDir**, read and written directly with **org.json**. No database, no ORM, no schema migrations. For a ledger used by a single shop owner on a single phone, Room's benefits (indexed queries, joins, migration tooling) do not apply: the whole dataset for a small shop fits in memory and serializes to disk in milliseconds. A flat JSON file is also trivially inspectable and portable, the same property that makes CSV export a direct, one pass transformation of the same data.

Decision 3: Hand rolled Drive backup, not the Drive Android SDK

Cloud backup is real but optional. Rather than pulling in Google's official Drive client, which drags in Play Services dependencies that bloat the APK and complicate offline first behavior, Transpent uses **AccountManager** for the OAuth token and raw **HttpURLConnection** multipart requests directly against the Drive v3 REST API. It is more code to write by hand, but it keeps the app's total dependency count at exactly two, AndroidX AppCompat and Material Components, and keeps the local only path from ever touching Google Play Services.

Decision 4: “Continue Locally” by default, not forced sign in

This one came directly out of the app's own fix log: early builds could not be entered on an emulator or device without first completing Google account selection. The fix was a Continue Locally button placed directly beneath Google sign in, opening the app immediately in local only mode; Drive backup now asks for sign in only when the owner actually taps it, from the Export screen. It is a small UI change with a real product consequence: a shop owner without a configured Google account, or without signal at the counter, is never blocked from using the app.

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The Build Story

Origin

Transpent began as a narrow, practical tool: a way for small shop owners and general stores to track credit and debt without a paper register. The first version, built through Google Antigravity, used a stark, high contrast Neobrutalist visual style chosen deliberately for maximum legibility: bold borders, flat blocks of color, no ambiguity about what was tappable.

As the feature set grew (item level payments, bill photo attachments, Drive backup, a statistics view), the Neobrutalist look stopped matching the app's ambition. A shop owner trusting an app with a customer's running balance needed it to look dependable, not just legible. The interface was rebuilt around Material Design 3: soft rounded cards, a contextual green/blue palette that carries real meaning (green for money coming in from customers, blue for money going out to suppliers), and a smooth pill switcher dashboard, without giving up the zero dependency, offline first foundation underneath.

Production hardening

The app's own fix log (**issues.md**) captures a single concentrated push that turned a debug only prototype into a distributable release: a signed keystore and release build type, a cleaned up **.gitignore** so build caches and local secrets stopped appearing in **git status**, a fix for card text that was clipping inside fixed height layouts, the “Continue Locally” bypass, and a first push of the finished repository to GitHub.

Date	Milestone	Status
Early build	Bespoke app built via Google Antigravity; Neobrutalist high contrast MVP; debug APK only	Done
May 14, 2026	Release keystore + signing config, .gitignore cleanup, card text clipping fix, “Continue Locally” added, repo pushed to GitHub	Done
May 14, 2026	Signed release APK installed & verified on an Android emulator	Done
Ongoing	Material Design 3 rebrand (green/blue), CSV export (single/split), Drive backup and restore wired end to end	Done
Jul 8, 2026	Real device verification with a live Google sign in; Drive backup auth path confirmed working outside the emulator	Done
Next	Search/filter polish, date range history filter, multi currency, Play Store release, home screen widget	Planne d

Real numbers

1 Java source file (MainActivity.java)	**8** screens: login plus 7 dashboard tabs	**9** modal dialogs for entries & payments
2 runtime dependencies (AppCompat, Material)	**18** hand authored vector icons	**24** min SDK (Android 7.0)
36 target SDK (Android 16)	**11** columns in every CSV export row	**4** data classes: Store, Party, Entry, Product

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The AI Collaboration Account

This project was built with real AI assistance across two different tools, and the nature of that collaboration is worth being specific about, because it was not "the AI wrote the app."

Ideation and product decisions: human

Every product decision in Transpent traces back to a real, unglamorous problem: small shop owners keeping credit and debt in a paper khata, with no digital fallback if the notebook is lost, damaged, or simply hard to read. The decision to keep the app fully usable without a Google account, to store data locally instead of on a server no one at the shop could administer, and to give Customers and Suppliers visually distinct identities so an owner never confuses who owes whom: those calls came from Vinny, not from a suggestion.

Implementation: Google Antigravity + Claude Code

Google Antigravity, Google's Gemini powered IDE, built the first working version of the app, the original Neobrutalist MVP, and later carried out the Material Design 3 redesign: the green/blue contextual palette, the pill switcher dashboard, the feature grid, and the supporting dialogs.

Claude Code, Anthropic's CLI coding agent, handled the release engineering pass that turned a debug only prototype into a distributable app: setting up the release keystore and signing config, cleaning **.gitignore** so build caches and local secrets stopped leaking into version control, diagnosing and fixing a card layout bug where 19sp to 22sp text was clipped inside fixed 48dp height cards, adding the "Continue Locally" bypass, installing the signed release build on an emulator to confirm it launched cleanly, and pushing the finished repository to GitHub.

> *The app's fix log reads like a running commit journal: Issue, Status, Action, one entry per problem, exactly the record an agentic coding tool produces while working a punch list, not a log of an AI choosing what to build.*

What the AI excelled at

- Turning a rough UI direction into working Material Components code
- Diagnosing a specific layout bug from a description and fixing it directly
- Handling the unglamorous parts of Android release engineering: keystores, signing configs, .gitignore, Gradle tasks, correctly on the first real attempt
- Verifying its own work by installing the signed build on an emulator rather than assuming it was correct

What it did not do

- Decide the app should be usable without a Google account
- Judge whether a color coded Customer / Supplier split would make sense to a shop owner
- Choose to store data as a local JSON file instead of a database
- Decide what belongs on the roadmap next

What's Next

Confirmed items pulled directly from the app's own roadmap and current source, not invented filler.

Chart on Statistics	Statistics currently shows real computed totals in two theme colored cards; a bar or line visualization is the natural next step.
In app bill viewer	Bill photos already attach via camera or gallery; there is no browsable in app viewer for them yet.
Date range filter on History	Filter entries by date range, not just by Customers / Suppliers.
Multi currency support	Ledger currently assumes a single currency (Rs).
Google Play Store release	Currently a signed APK distributed via GitHub Releases.
Home screen widget	Quick glance at pending balance without opening the app.

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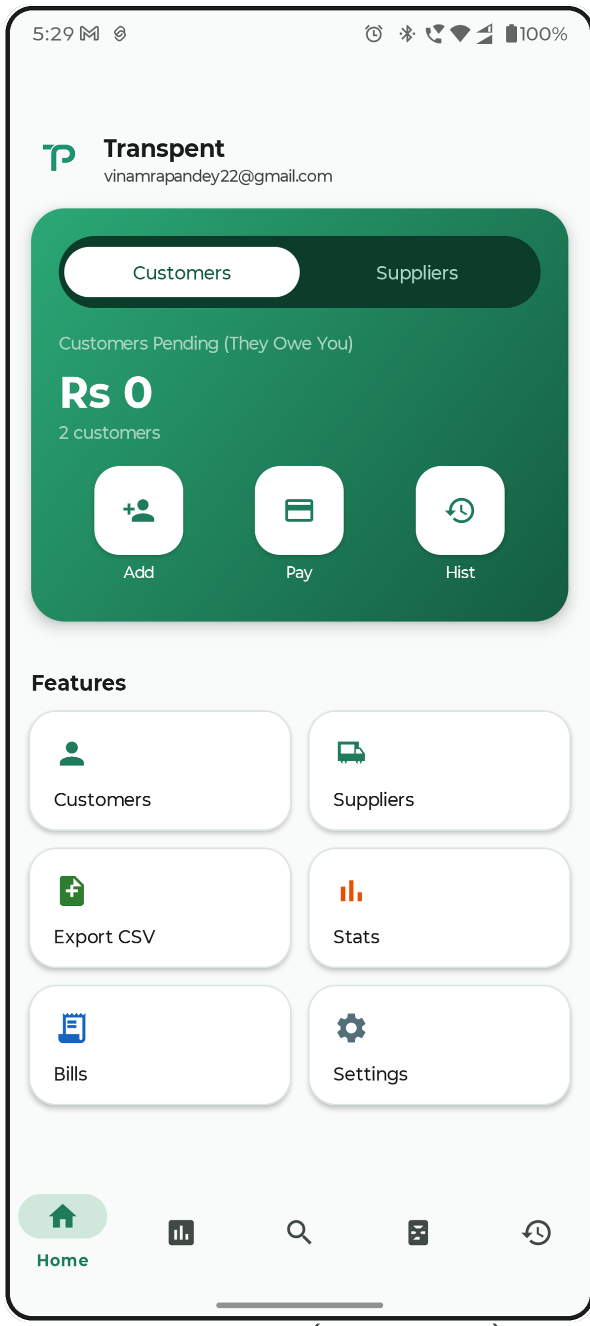
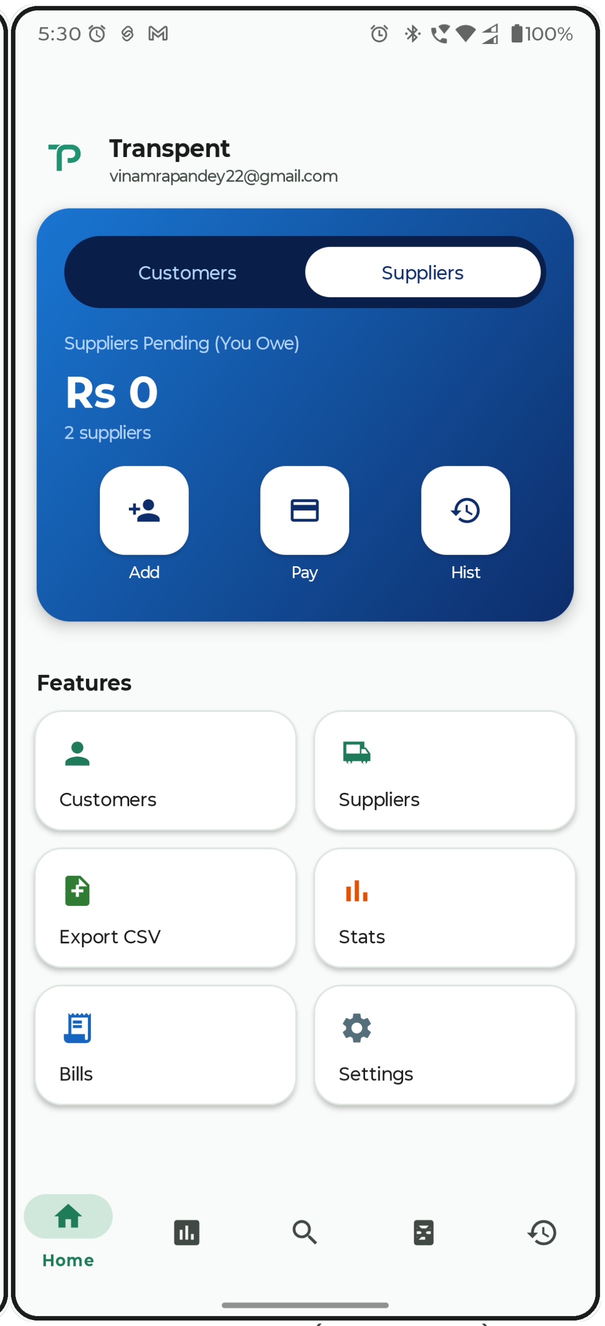
App Screenshots

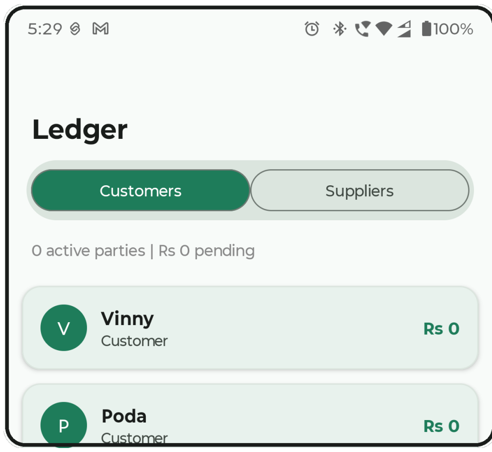
Captured on device, July 8, 2026, the same build described throughout this document, including a live Google sign in confirming the Drive backup auth path outside the emulator.



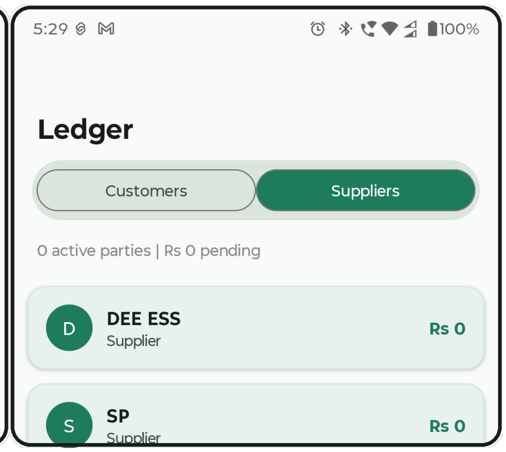
Home: Customers (emerald green)



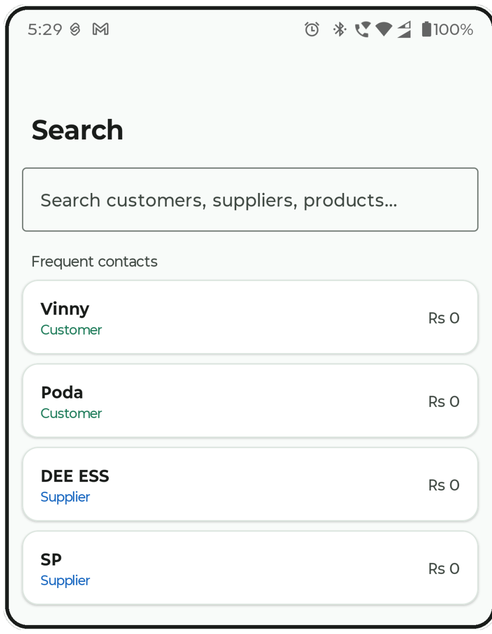
Home: Suppliers (sapphire blue)



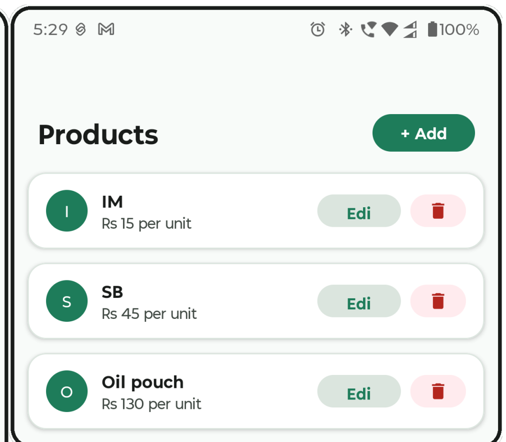
Ledger: Customers, with FAB



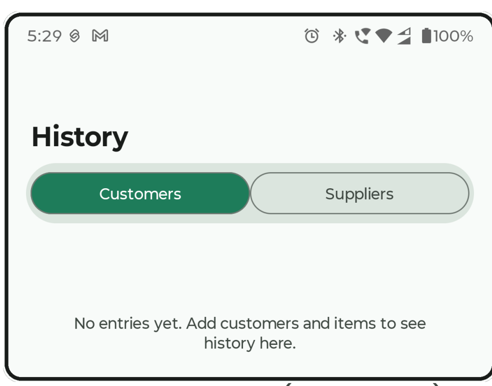
Ledger: Suppliers



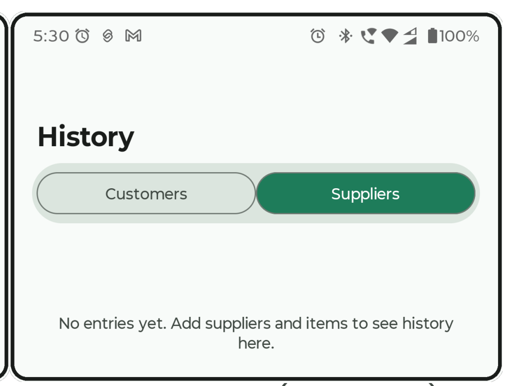
Search: frequent contacts



Products catalog



History: Customers (empty state)



History: Suppliers (empty state)

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Resources

Links

Resource	Value
Repository	github.com/vinamrapandey/Transpent
Releases (signed APK)	github.com/vinamrapandey/Transpent/releases
Package	com.transpent.app
License	MIT
Contact	vinamrapandey22@gmail.com

> *Transpent is a working answer to a small, specific problem: built, hardened, and still in active development.*